



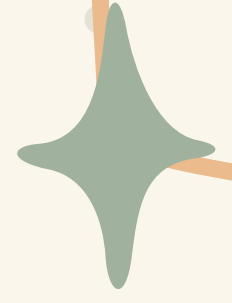
THE CHAI STRAINER AND THE SALT PAN





KABIR AND ANYA'S RAINY DAY CHALLENGE

After heavy rain, Kabir and Anya's cricket ground was a muddy mess! Their water bucket was mixed with dirt and sand. Can you help them get clean water, clean sand, and salt back?

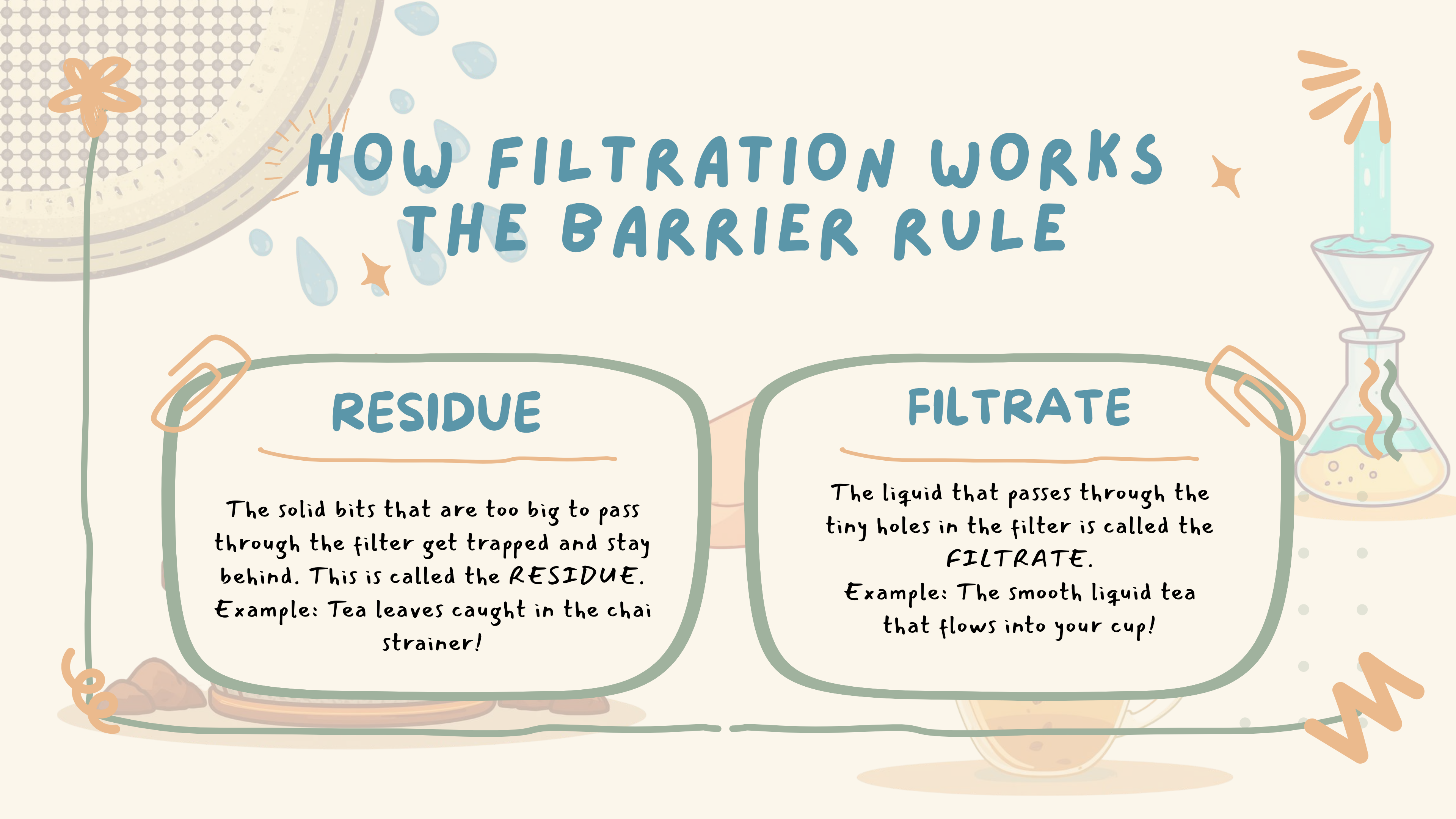


WHAT IS A MIXTURE?

When two or more things are mixed but
can still be separated!

Examples: sugar in water, sand in
water, tea with leaves

Soluble = can dissolve • Insoluble =
cannot dissolve



HOW FILTRATION WORKS THE BARRIER RULE

RESIDUE

The solid bits that are too big to pass through the filter get trapped and stay behind. This is called the *RESIDUE*.
Example: Tea leaves caught in the chai strainer!

FILTRATE

The liquid that passes through the tiny holes in the filter is called the *FILTRATE*.
Example: The smooth liquid tea that flows into your cup!



INDIAN EXAMPLE: THE CHAI STRAINER



"Chai-ni" (tea strainer) lets liquid tea pass through but catches the tea leaves behind!

 Residue = Tea leaves (trapped in strainer) |

Filtrate = Liquid tea (passes through)



HOW EVAPORATION WORKS!

When water is heated, it turns into steam and floats away — leaving solids behind! Dissolved solids like salt are too tiny for a filter, so we use heat instead. The water disappears, and only the solid remains.

⚠ Safety Note: Always do heating experiments with adult supervision!



INDIAN EXAMPLE: THE SALT PAN

In Gujarat and Tamil Nadu, sunlight evaporates seawater in vast salt pans. As the water disappears into steam, salt crystals are left behind — this is natural evaporation! Salt farmers then collect the white salt crystals from the pans.





WHEN WE NEED BOTH!

STEP 1: FILTRATION

Pour the sand + salt + water mixture through a filter cloth.

✓ Sand gets **TRAPPED** — this is the Residue!

💧 Salt water passes through — this is the Filtrate!

STEP 2: EVAPORATION

Heat the salt water (filtrate) gently in a pan.

☁ Water turns to steam and disappears!

✓ Salt **CRYSTALS** are left behind — just like the Salt Pans of Gujarat!



GUIDED PRACTICE: FILTERING RIVER WATER



Kabir folds a piece of Suti Kapda (white cotton cloth) and places it over an empty bucket. He slowly pours muddy river water through the cloth and watches carefully...

What gets **TRAPPED** in the cloth?

- Silt (fine mud particles)
- Dirt and small pebbles
- Tiny bits of leaves and debris

What **PASSES THROUGH** the cloth?

- Clear, cleaner water (the **Filtrate**!)

The cotton cloth acts like a filter — its tiny holes let water through but block bigger solid particles.

The trapped dirt is the **Residue**. The clean water collected below is the **Filtrate**!

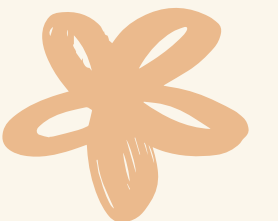




SALT HARVESTING AT RANN OF KUTCH

Salt farmers at the Rann of Kutch collect salt after the bright sun evaporates seawater. Step 1: Seawater flows into wide, shallow salt pans. Step 2: The hot sun heats the water day after day. Step 3: Water turns into steam and rises into the air!

Step 4: White salt crystals are left behind in the pans. Step 5: Farmers rake and collect the salt. This is natural evaporation at work! The dissolved salt cannot escape as steam — only water can. Gujarat's Rann of Kutch is one of the world's largest salt deserts!



MATCH THE MIXTURE TO THE METHOD!



FILTER 
PEBBLES + WATER
TURMERIC + WATER

EVAPORATE 
SUGAR + WATER
SALT + WATER



SUMMARY: REMEMBER THESE RULES!

FILTER IT!



Insoluble = Filter it!
Big particles get trapped.
Some mixtures need BOTH steps — filter first, then evaporate!
Key Words: Mixture • Filtration • Residue • Filtrate

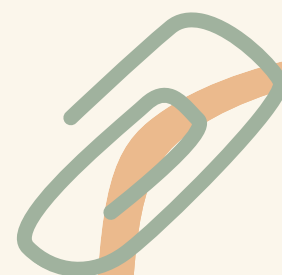
EVAPORATE IT!



Soluble = Evaporate it!
Liquid disappears, solid stays.
Salt is soluble — you cannot catch it with a filter!
Key Words: Soluble • Insoluble • Evaporation • Crystals



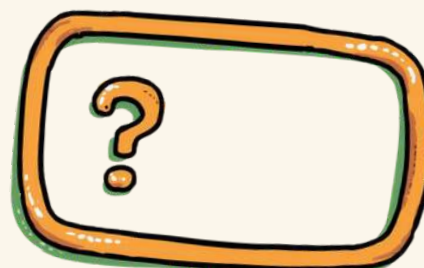
QUICK QUIZ TIME!



QUESTION 1

What does a filter catch? What is the solid left behind called?

RESIDUE! ✓



QUESTION 2



What happens to water during evaporation? Can salt be removed by filtering?

STEAM! NO! ✓





YOUR HOME MISSION!

WATCH YOUR PARENTS MAKE CHAI (TEA) OR FILTER WATER
AT HOME. BE A SCIENCE DETECTIVE! 🔍

- ✓ SPOT THE RESIDUE — WHAT GETS TRAPPED BEHIND? (TEA LEAVES, DIRT, SILT)
- ✓ SPOT THE FILTRATE — WHAT PASSES THROUGH? (LIQUID TEA, CLEAN WATER)
- ✓ CAN YOU FIND EVAPORATION HAPPENING TOO? (STEAM FROM THE POT!)
- ★ SHARE WHAT YOU DISCOVER IN OUR NEXT CLASS — WE CAN'T WAIT TO HEAR YOUR FINDINGS!



THANK
YOU!

KEEP EXPLORING SCIENCE!